

TEJAS PANDYA

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EDUCATION

NEW YORK UNIVERSITY, TANDON SCHOOL OF ENGINEERING , Brooklyn, NY	Expected 05/27
Master of Science in Financial Engineering GPA: 3.8/4.0	
Graduate Teaching Assistant: Financial Economics; C++ Financial Software Laboratory; Value Investing	
SVKM'S NMIMS, SCHOOL OF BUSINESS MANAGEMENT , Mumbai, India	04/22
Master of Business Administration, Concentration in Finance GPA: 3.3/4.0	
VELLORE INSTITUTE OF TECHNOLOGY , Vellore, India	06/20
Bachelor of Technology in Mechanical Engineering GPA: 8.9/10.0	

PROGRAMMING / TECHNICAL SKILLS / CERTIFICATIONS

Programming: Python, SQL, C/C++ (C++20, STL), Bash, LaTeX/Typst
Libraries & Frameworks: NumPy, Pandas, Polars, SciPy, scikit-learn, statsmodels, LightGBM, XGBoost, PyTorch, pytest
Data & Tools: Linux/Unix, Git, CMake, Streamlit, Shiny, Excel, Bloomberg Terminal, Tableau, Power BI
Certifications: CFA Level I; C++ Programming for Financial Engineering (Baruch College); Bloomberg Market Concepts; CFI FMVA

COURSEWORK HIGHLIGHTS

Risk, Valuation & Derivatives: Financial Risk Management, Valuation for Financial Engineering, Quantitative Portfolio Management, Options Pricing & Stochastic Calculus, Fixed Income Securities and Interest Rate Derivatives
Quantitative Methods & Computing: Financial Econometrics, Machine Learning in Financial Engineering, Quantitative Methods in Finance, Financial Computing, Applied Numerical Methods, Data Structures and Algorithms

EXPERIENCE

NEW YORK UNIVERSITY, TANDON SCHOOL OF ENGINEERING , Brooklyn, NY	05/26 - 08/26
<i>Quantitative Researcher, QWIM Industry Capstone</i>	
- Lifted Sharpe from 0.37 to 0.55 against a 60/40 benchmark over 1999-2025 as one of five on the QWIM capstone, pairing equal-risk-contribution weighting with hierarchical budgeting on Conditional Drawdown-at-Risk and blending regime-specific covariance by one-month-ahead probability - Compressed annualised volatility from 11.5% to 6.3%, maximum drawdown from 37% to 24% and Conditional Drawdown-at-Risk from 0.250 to 0.169 against the same 60/40 benchmark, accepting a marginally lower absolute return of 5.4% against 5.7% and turnover of 0.242 - Diagnosed thirteen environment-configuration defects behind a 278-file team pytest suite that reported success while executing zero tests on macOS, restoring 10,423 tests to green and documenting the 22 pre-existing failures rather than leaving them quietly reported as green	
NOMURA SERVICES INDIA PRIVATE LIMITED , Mumbai, India	08/22 - 07/25
<i>Senior Risk Analyst, Market Risk Infrastructure</i>	
- Owned the lifecycle of VaR and sensitivity models across equities, rates, FX and commodities, from daily production runs through backtesting; Python anomaly detection triaged exceptions to stale marks, erroneous data or booking errors, cutting investigation turnaround 40% and restoring SLAs - Delivered quarterly Basel 2.5 and FRTB stress-testing and capital submissions alongside front-office and regulatory teams, engineering rules-based validation over delta, vega and curvature sensitivities that reconciled fully against front-office pricing and caught anomalies pre-submission - Engineered Python and SQL pipelines across equities, rates, FX and commodities that aggregate, validate and reconcile daily P&L, positions and Level-1/Level-2 market data into the VaR and sensitivity dashboards, eliminating more than 50 hours of manual processing every month - Led a three-person workstream on US-region end-of-day risk reporting, cutting overnight batch runtime 35% through SQL query optimisation and job scheduling, and built the VaR, P&L and exception dashboards that senior stakeholders read each morning before the US open	

ACADEMIC PROJECTS

CROSS-SECTIONAL EQUITY ALPHA RESEARCH	07/26
<i>Python, LightGBM, statsmodels</i> github.com/6ixE11even/equity-xs-alpha	
- Constructed eight cross-sectional signals across 495 S&P 500 names, scored by Spearman rank information coefficient with Newey-West t-statistics under Benjamini-Hochberg false-discovery control; only Amihud illiquidity survived at 5.32, while the low-volatility and lottery signals inverted - Pulled 15,575 SEC filings from EDGAR at the published rate limit, scoring Item 1A risk factors against the Loughran-McDonald finance dictionary rather than a general-purpose lexicon; 93% were scored on Item 1A itself and the rest fell back to MD&A or the whole document	
TREASURY-CURVE SWAP VALUATION & HISTORICAL COUNTERPARTY RISK	05/26
<i>Python, NumPy, SciPy</i> github.com/6ixE11even/sofr-swap-pnl-attribution	
- Bootstrapped discount curves from FRED Treasury constant-maturity par yields rather than licensed SOFR fixings, marking a \$255M ten-swap 2-30 year book and decomposing daily P&L into carry, roll-down and orthogonal level, slope and curvature moves reconciling to the total within 1e-6 - Priced historical counterparty CVA from 20,000 paired exposure paths over 8,478 days of joint rate and credit history, measuring wrong-way risk at \$89,553, 16.9% above a decoupled baseline sharing its curve paths bit-for-bit, at a \$2.35M peak expected exposure, with no collateral agreement	
OPTIONS PRICING ENGINE - VANILLA, AMERICAN AND BARRIER	10/25
<i>C++20, CMake, CTest</i> github.com/6ixE11even/options-pricing-engine	
- Implemented three independent valuation methods behind one interface: closed-form Black-Scholes with analytic greeks, Monte Carlo with exact terminal sampling, and a Crank-Nicolson solver using projected SOR for American early exercise, with barriers as domain boundaries - Cross-validated Black-Scholes, Monte Carlo and Crank-Nicolson to within 0.6% on an at-the-money European call, pinning that agreement as regression tests at \$0.02 and \$0.06 tolerances, recovering a 0.51 early-exercise premium and documenting the discrete-monitoring gap on barriers	